

Data Science Project Proposals Based on Bangladesh Weather Dataset

1. Introduction

Weather plays a major role in agriculture, transport, and daily life in Bangladesh, and analyzing weather data can help predict future conditions and support better planning. For this course, I collected a weather dataset for Bangladesh through data scraping and cleaned it for analysis. This report proposes four Data Science projects that can be built using this dataset, covering classification, regression, and clustering techniques.

2. Dataset Description

The dataset contains about 10,000 records with 20 features collected from multiple weather stations across Bangladesh. It includes station information, date and time, season, temperature, rainfall, humidity, wind speed, and surface pressure. The main prediction target is Target_Rain_Tomorrow, which shows whether it will rain the next day.

Feature	Description
Station_Name / Division	Weather station and division
Season	Season of observation
Current_Temp_C	Current temperature
Rainfall_mm	Rainfall amount
Humidity_Morning/Evening_Pct	Morning / evening humidity
Wind_Speed_kmh	Wind speed
Surface_Pressure_hPa	Surface pressure
Target_Rain_Tomorrow	Rain / No Rain target

3. Proposed Projects

Proposal 1: Rain Tomorrow Prediction

Objective: This project will predict whether it will rain the next day using Target_Rain_Tomorrow as the target.

Features & Model: Features: Current_Temp_C, Temp_Max_C, Temp_Min_C, Rainfall_mm, Humidity_Morning_Pct, Humidity_Evening_Pct, Wind_Speed_kmh, Surface_Pressure_hPa, Rainfall_Lag1_mm, Season. Model: Logistic Regression / Random Forest Classifier.

Expected Output: The model will output Rain / No Rain and will be evaluated using accuracy, precision, recall, F1-score, and a confusion matrix.

Application: Helps in short-term rain forecasting for daily planning.

Proposal 2: Rainfall Amount Prediction

Objective: This project will predict the numerical rainfall amount (Rainfall_mm) for a given day using weather conditions.

Features & Model: Features: Current_Temp_C, Humidity_Morning_Pct, Humidity_Evening_Pct, Wind_Speed_kmh, Surface_Pressure_hPa, Rainfall_Lag1_mm, Season. Model: Linear Regression / Random Forest Regressor.

Expected Output: The model will output predicted rainfall in mm and will be evaluated using MAE, RMSE, R^2 score, and an actual-vs-predicted graph.

Application: Supports flood risk assessment and agricultural planning in Bangladesh.

Proposal 3: Temperature Prediction

Objective: This project will predict Current_Temp_C using other weather features, while avoiding features that directly cause target leakage.

Features & Model: Features: Humidity_Morning_Pct, Humidity_Evening_Pct, Wind_Speed_kmh, Surface_Pressure_hPa, Season, Division. Model: Linear Regression / Random Forest Regressor.

Expected Output: The model will output predicted temperature and will be evaluated using MAE, RMSE, R^2 score, and an actual-vs-predicted visualization.

Application: Useful for short-term temperature forecasting.

Proposal 4: Weather Pattern / Station Clustering

Objective: This project will group weather stations or observations with similar weather patterns using unsupervised learning.

Features & Model: Features: Current_Temp_C, Rainfall_mm, Humidity_Morning_Pct, Humidity_Evening_Pct, Wind_Speed_kmh, Surface_Pressure_hPa (scaled before clustering). Model: K-Means Clustering.

Expected Output: The output will be weather clusters showing groups of stations/observations with similar characteristics, along with cluster visualizations and comparisons.

Application: Helps understand regional and seasonal weather patterns across Bangladesh.

4. Proposal Comparison

Proposal	Task	Model	Main Output
1	Rain Prediction	Classification	Rain / No Rain
2	Rainfall Prediction	Regression	Rainfall in mm
3	Temperature Prediction	Regression	Temperature
4	Weather Pattern Analysis	K-Means	Weather Clusters

5. Expected Overall Outcome

The four proposed projects will together demonstrate classification, regression, and clustering techniques using the same cleaned Bangladesh weather dataset. Each project will include model training, evaluation, and visualization of results. Overall, this will show how one dataset can support multiple Data Science applications.

6. Conclusion

The collected Bangladesh weather dataset is suitable for multiple Data Science and Machine Learning tasks. The four proposed projects will provide useful, practical insights into rainfall, temperature, and regional weather patterns, and will help build strong hands-on experience with classification, regression, and clustering methods.